

AYE
TECH HUB

• PLC PROGRAMMING • LESSON 04

Ladder Logic

Fundamentals

Lesson 04 of 30 | PLC Programming Program

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- What Ladder Logic is and why it mirrors relay control circuits
- The 7 basic Ladder Logic symbols — contacts, coils, power rails
- XIC, XIO, OTE: the three instructions that run 80% of all programs
- Classic Start/Stop motor control rung — step by step
- Seal-in (latching) circuits and why they are essential

What Is Ladder Logic?

Ladder Logic is the most widely used PLC programming language in the world. It was designed in the 1970s to look exactly like the relay control circuits it was replacing — so that electrical technicians could read and write PLC programs without needing computer programming skills. The name comes from the appearance of the program: two vertical **power rails** (like the sides of a ladder) connected by horizontal **rungs** (like the steps).

Each rung is a logic equation. The PLC evaluates every rung from top to bottom, left to right, during every **scan cycle** (typically 1–50 ms). A rung outputs TRUE (energises the coil) only when all the logic conditions on that rung evaluate to TRUE — just like current flowing through a relay circuit.

The 7 Basic Ladder Logic Symbols

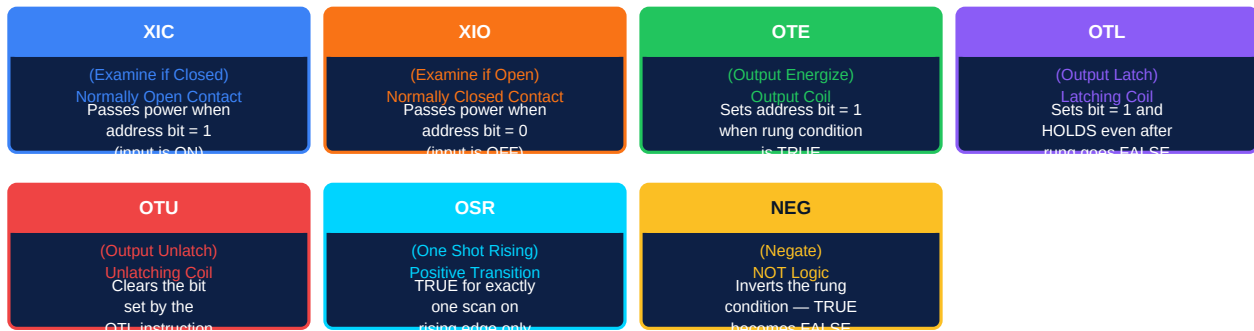


Figure 1 — Basic Ladder Logic Instruction Symbols

Symbol	IEC Name	Siemens (S7)	Allen-Bradley	Mitsubishi	Description
NO Contact (XIC)	LD (Load)	I0.0 — —	XIC — —	X000 — —	Passes power when the addressed bit is 1 (ON)
NC Contact (XIO)	LDN (Load NOT)	I0.0 — / —	XIO — / —	X000 — / —	Passes power when the addressed bit is 0 (OFF)
Output Coil (OTE)	OUT	Q0.0 —()—	OTE —()—	Y000 —()—	Energises (sets to 1) when rung condition is TRUE
Latch Coil (OTL)	SET	Q0.0 —(S)—	OTL —(L)—	—[SET]—	Latches ON; stays ON even if rung goes FALSE
Unlatch Coil (OTU)	RESET	Q0.0 —(R)—	OTU —(U)—	—[RST]—	Clears the bit set by the latch coil

IEC 61131-3 standard: Ladder Logic (LD) is one of five official PLC programming languages defined by IEC 61131-3. The others are: FBD (Function Block Diagram), ST (Structured Text), IL (Instruction List), and SFC (Sequential Function Chart). All modern PLCs support all five, but Ladder remains the most common for discrete control.

The Classic Start/Stop Motor Control Circuit

The start/stop motor control circuit is the "Hello World" of PLC programming. Every PLC programmer must understand it completely before moving to more complex logic. It demonstrates the core concepts of **series logic (AND)**, normally-closed contacts, and the fundamental challenge of latching — keeping the motor running after the start button is released.

Step 1: Basic Start/Stop (No Seal-In)

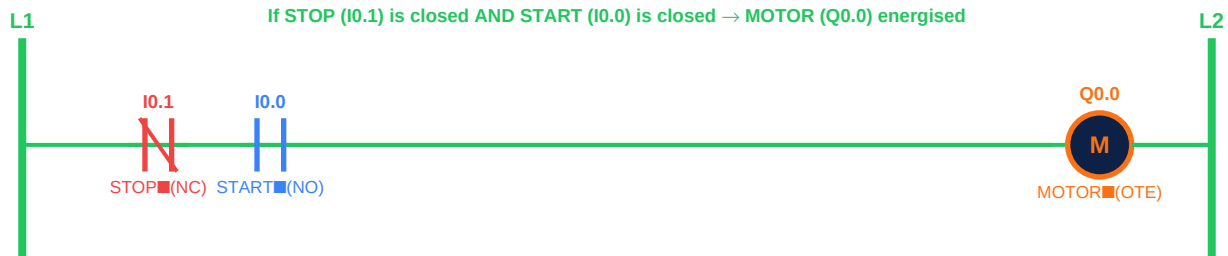


Figure 2 — Basic Start/Stop Motor Control Rung (No Seal-In)

Problem with this circuit: As soon as you release the START button, I0.0 goes back to 0, the rung goes FALSE, and the motor stops. This is correct for a momentary "jog" function but not for a "start and keep running" application. The solution is the seal-in circuit.

Step 2: Seal-In (Latching) Circuit

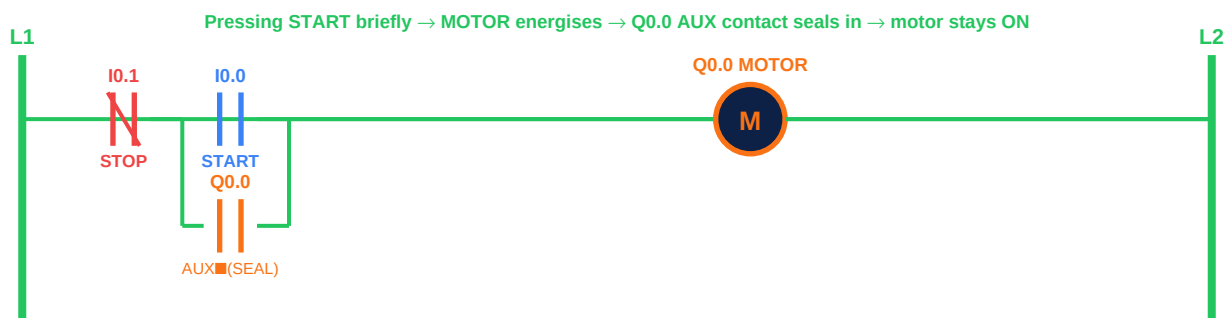


Figure 3 — Seal-In (Latching) Circuit: START+STOP with Auxiliary Contact

How the seal-in works: When START (I0.0) is pressed, the motor coil (Q0.0) energises. Q0.0 also has an auxiliary contact on the same rung in parallel with START. Once the motor is running, Q0.0 closes its own auxiliary contact — this "seals in" the rung, keeping power flowing even after START is released. Only pressing STOP (I0.1, normally-closed) breaks the circuit.

Series, Parallel Logic and Best Practices

Logic Type	Ladder Layout	Boolean	Real Example
Series (AND)	Contacts connected left-to-right on same rung — — — — — — —()—	Output = A AND B AND C	Motor runs only when: Guard_closed AND Pressure_OK AND Start_pressed
Parallel (OR)	Contacts connected on separate branches — — (branch 1) — — (branch 2) both connecting to same output	Output = A OR B	Alarm sounds if: High_Temp OR Low_Pressure OR Emergency_Stop
Series + Parallel	Mix of branches and series contacts — — —(— —(branch))—	Output = A AND (B OR C)	Motor runs: Safety_OK AND (Start OR Auto_Start)
Normally Closed Logic	NC contact (XIO) in series with output — / — —()—	Output = NOT A	Allow if NOT Fault_active — safety interlock using NC

Ladder Logic Best Practices

Rule	Correct Practice	Why It Matters
Power flow is left to right	Inputs (contacts) always on the left; outputs (coils) always on the right	Any other arrangement confuses the scan and causes logic errors
One output address per program (usually)	Each output bit (Q0.0, etc.) should appear as an OTE coil only once	Multiple OTE coils on the same address: last one wins — earlier rungs are overridden
Use OTL/OTU pairs for latching	Use separate Set (OTL) and Reset (OTU) rungs instead of complex seal-ins for clarity	Easier to troubleshoot; less risk of unintended latching
Comment every rung	Add a rung comment explaining what each rung does	Unmarked programs are unreadable for maintenance 2 years later
Address consistency	Use consistent naming: I0.0 = Start, I0.1 = Stop — document in I/O list	Makes fault-finding possible; critical for handover and maintenance
Keep rungs simple	Maximum 5–6 contacts per rung; split complex logic into multiple rungs	Complex rungs are hard to debug; simple rungs are easy to trace
Test in simulation first	Use offline simulation mode before downloading to hardware	Prevents damage to equipment from logic errors; required for safety systems

Ladder Logic Practice Exercises

Exercise	Write a Rung That...	Hint
1 — AND gate	Turns on output Q0.0 only when I0.0 AND I0.1 are both ON	Two XIC contacts in series → one OTE coil
2 — OR gate	Turns on Q0.0 if either I0.0 OR I0.1 is ON	Two XIC contacts in parallel → one OTE coil
3 — NOT gate	Turns on Q0.0 only when I0.0 is NOT active	One XIO (normally-closed) contact → OTE coil

4 — Start/Stop seal-in	Runs motor Q0.0 when I0.0 is pressed; stops when I0.1 is pressed	NC stop + (NO start Q0.0 aux) → OTE motor coil
5 — Emergency stop	Stops all outputs immediately when I0.7 (E-STOP) is pressed	Use I0.7 as XIO (NC) contact in series with every output rung

What comes next — PLC-005: Timer and Counter Instructions — TON (Timer On Delay), TOF (Timer Off Delay), RTO (Retentive Timer), CTU/CTD (Count Up/Count Down) — the four instructions that make PLCs handle time-based and quantity-based control automatically.